Practice workbook · Identify parts of the eye and state their functions

Naming labelled eye structures, choosing structures from descriptions, matching parts to jobs, and describing roles of cornea, lens, retina, fovea, iris and the blind spot.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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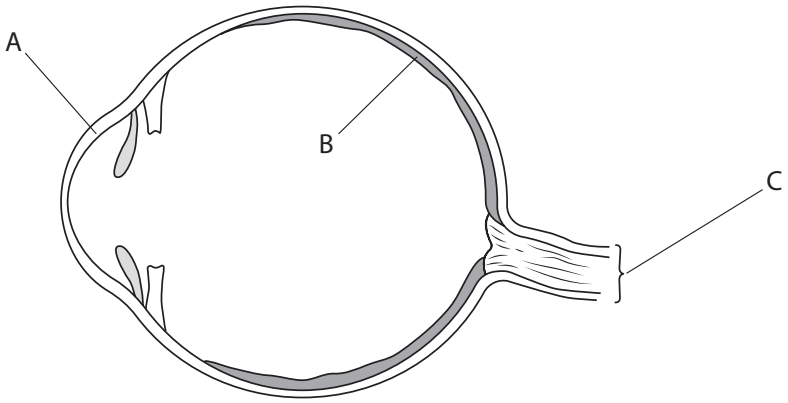
Type ID: HBI-IDENTIFY-EYE-PARTS-AND-ROLES

1. Question 2 · part a.ii · 3 marks

June 2025 Paper 1R

Worked-example retry · Attempt part a.ii only.

2 The diagram shows a section through the eye with the lens missing.



(a) (i) Complete the diagram by drawing in the lens and suspensory ligaments as they would appear when the eye is looking at a near object.

(2)

(ii) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(b) A person is looking at a near object.

Describe the changes that occur in the eye if the person then looks at a distant object.

(4)

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(Total for Question 2 = 9 marks)

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1. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

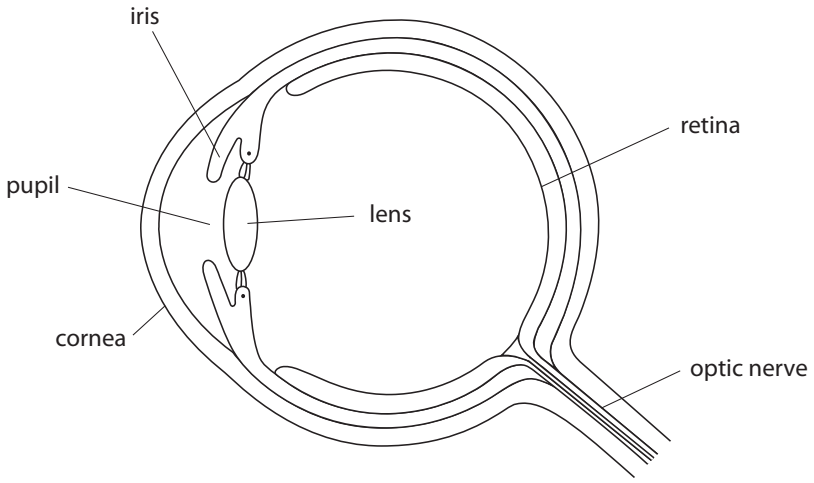
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2. Question 3 · part a · 1 mark

June 2019 Paper 1

Further practice · Attempt part a only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part a · 1 mark

June 2019 Paper 1

Further practice · Attempt part a only.

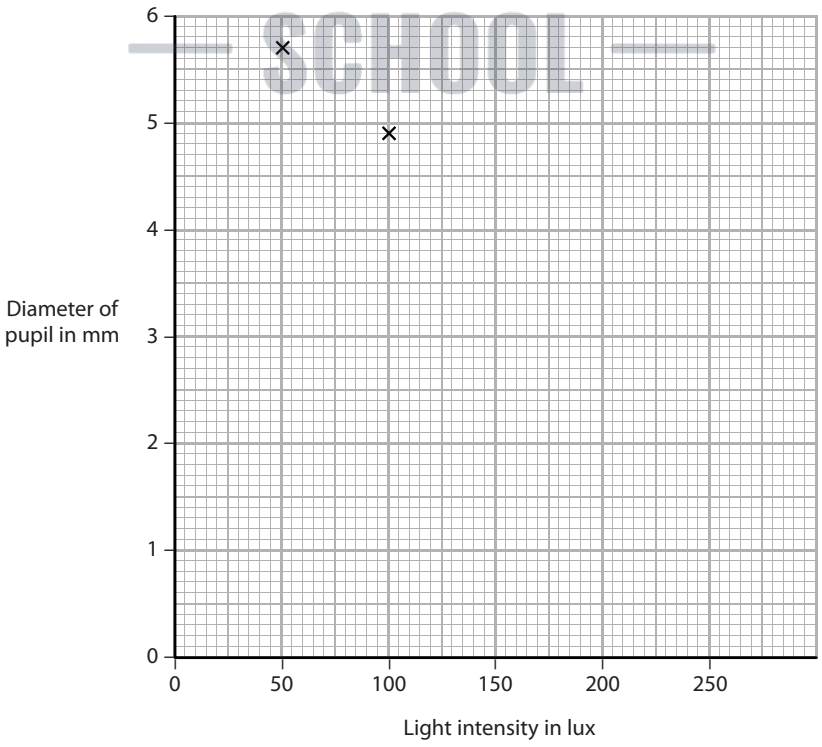
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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2. Question 3 · part a · 1 mark · Working and answer

Working and answer

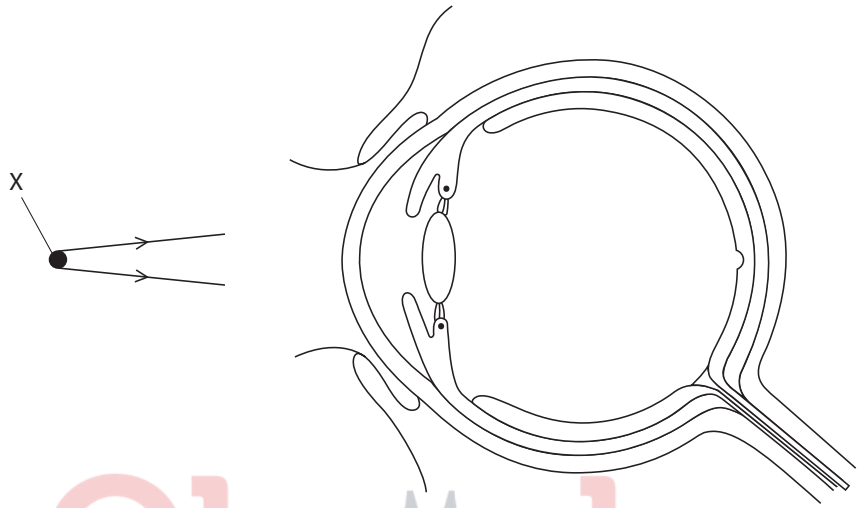
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3. Question 1 · part a · 5 marks

June 2019 Paper 2R

Further practice · Attempt part a only.

1 The diagram shows a section through an eye.



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(a) The table lists descriptions of parts of the eye.

Use the letters, A, B, C, D and E, to label the parts of the eye on the diagram.

(5)

Description of part	Letter
the layer that contains light sensitive cells	A
the muscles that change lens shape	B
the structure that controls the amount of light entering the eye	C
the structure that carries nerve impulses away from the eye	D
the structure that protects the eye from dust and grit	E

(b) A person looks at object X.

Complete the diagram by continuing the rays of light to show how the image of object X is formed on the retina.

(3)

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3. Question 1 · part a · 5 marks · Working and answer

Working and answer

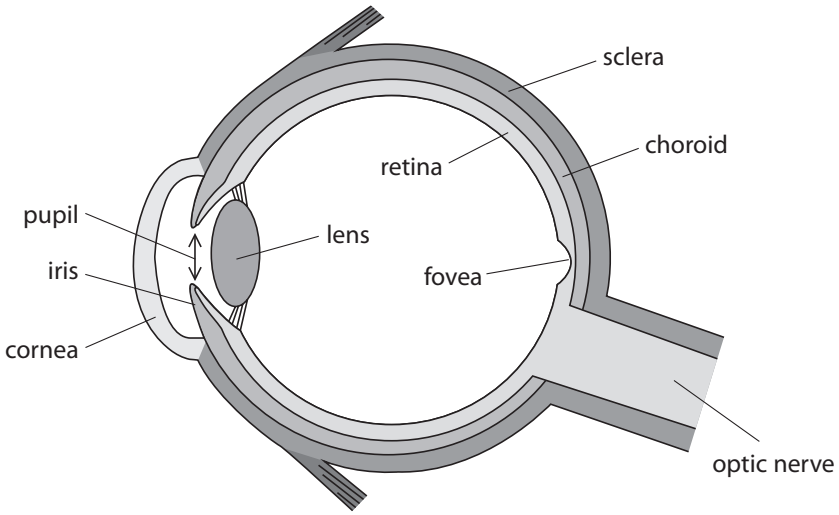
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4. Question 3 · part a.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.i only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye? **(1)**

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye. **(3)**

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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4. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

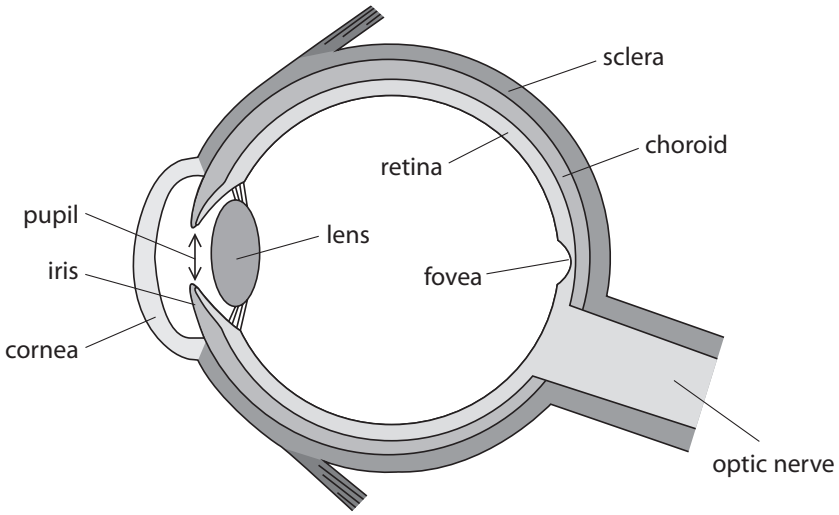
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5. Question 3 · part a.ii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye?

(1)

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye.

(3)

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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5. Question 3 · part a.ii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

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(b) The photomicrograph gives the thickness of the two outer layers of the eye.



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Estimate the thickness of the retina.

(1)

thickness of the retina = ... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

(2)

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(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

(2)

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(Total for Question 3 = 9 marks)

5. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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6. Question 6 · part a · 1 mark

June 2021 Paper 2

Further practice · Attempt part a only.

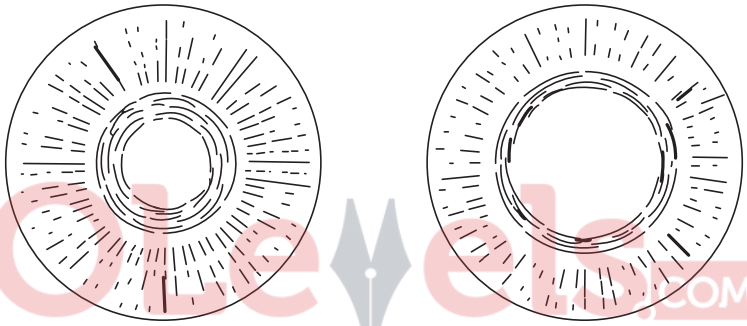
6 The human eye consists of many parts.

(a) Which part has the highest concentration of light sensitive cells?

(1)

- ☒ **A** blind spot
- ☒ **B** choroid
- ☒ **C** fovea
- ☒ **D** optic nerve

(b) The diagram shows the iris of an eye in two different light conditions, A and B.



light condition A

light condition B

(i) Describe what happens in the eye when the light condition changes from A to B.

(5)

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6. Question 6 · part a · 1 mark · Working and answer

Working and answer

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7. Question 8 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

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(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

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(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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7. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

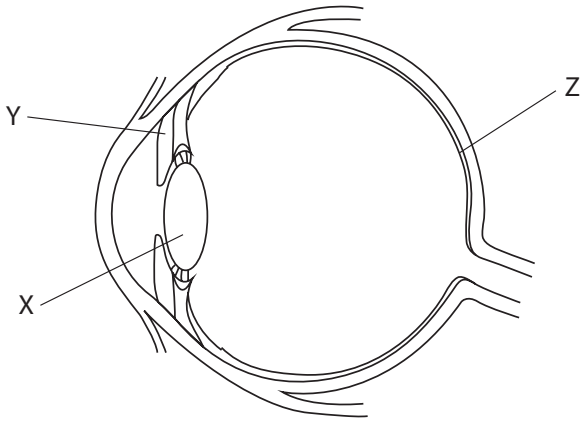
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8. Question 1 · part a.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
structure X	refracts light rays
structure Y	protects the eye from pathogens
structure Z	detects light rays
	controls the shape of the lens
	transmits electrical impulses to the brain
	controls the size of the pupil

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8. Question 1 · part a.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

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(ii) State one treatment for cataracts.

(1)

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(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

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(Total for Question 1 = 10 marks)

8. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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9. Question 6 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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9. Question 6 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

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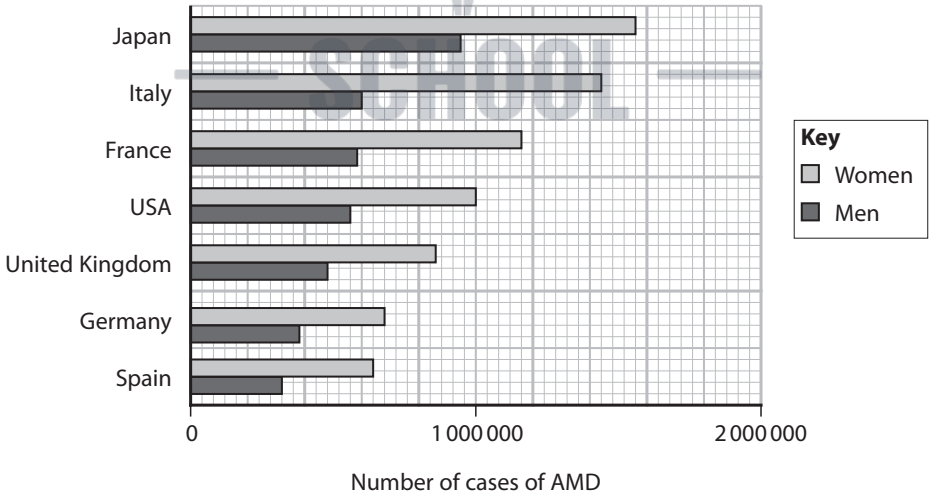
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.



(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

(2)

difference = ...

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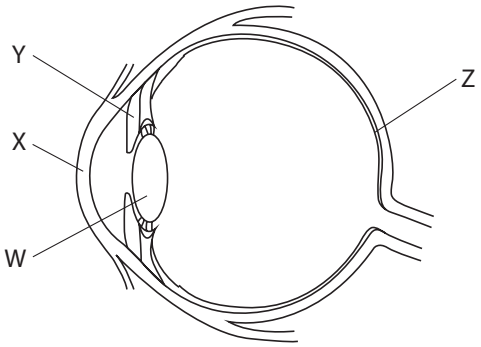
9. Question 6 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(ii) The diagram shows a human eye.

AMD affects the retina of the eye.



Which of these structures is affected by AMD?

(1)

- ☒ **A** structure W
- ☒ **B** structure X
- ☒ **C** structure Y
- ☒ **D** structure Z

(c) A deficiency of vitamin A can also cause blindness.

A genetically engineered rice product can be used to prevent deficiency of vitamin A.

Describe how genetic engineering is used to produce this rice product.

(3)

(Total for Question 6 = 11 marks)

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9. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

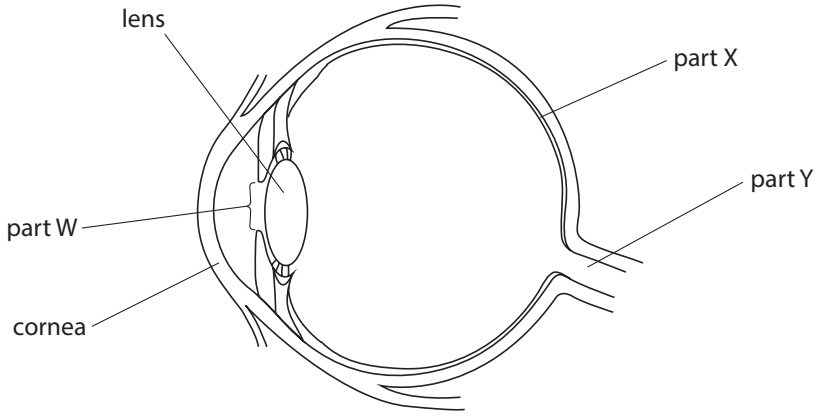
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10. Question 3 · part a.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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10. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

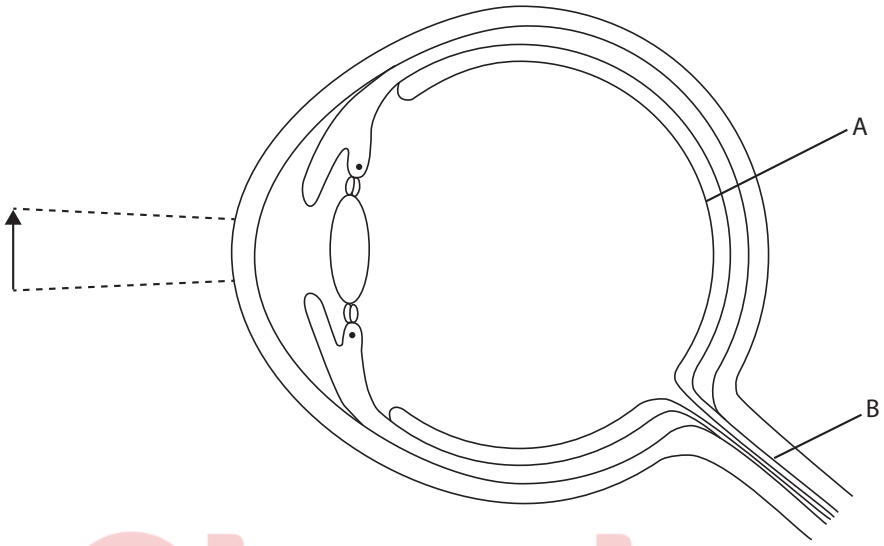
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11. Question 6 · part a.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.i only.

6 The diagram shows a section through the eye.



(a) (i) Describe the functions of structures A and B. **(4)**

A .....

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B .....

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(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

(4)

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11. Question 6 · part a.i · 4 marks · Working and answer

Working and answer

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12. Question 1 · part b · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

1 (a) The box lists some parts of the eye and the ear.

auditory nerve	ciliary body	cochlea	iris	lens	optic nerve
ossicles	pupil	retina	semi-circular canals	tympanum	

Complete the table using words from the box to give the missing information.

(6)

Description	Part
contains light-sensitive cells	
membrane that transfers vibrations to middle ear	
transmits impulses from the retina to the brain	
helps the body to balance	
contains muscle tissue that controls light entering the eye	
converts vibrations into nerve impulses	

(b) Explain what is meant by the blind spot in the eye.

(2)

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(Total for Question 1 = 8 marks)

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12. Question 1 · part b · 2 marks · Working and answer

Working and answer

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13. Question 1 · part a · 6 marks

June 2023 Paper 1R

Further practice · Attempt part a only.

1 (a) The box lists some parts of the eye and the ear.

auditory nerve	ciliary body	cochlea	iris	lens	optic nerve
ossicles	pupil	retina	semi-circular canals	tympanum	

Complete the table using words from the box to give the missing information.

(6)

Description	Part
contains light-sensitive cells	
membrane that transfers vibrations to middle ear	
transmits impulses from the retina to the brain	
helps the body to balance	
contains muscle tissue that controls light entering the eye	
converts vibrations into nerve impulses	

(b) Explain what is meant by the blind spot in the eye.

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(Total for Question 1 = 8 marks)

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13. Question 1 · part a · 6 marks · Working and answer

Working and answer

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14. Question 1 · part a · 7 marks

November 2023 Paper 2

Further practice · Attempt part a only.

1 (a) The box lists parts of the eye and parts of the ear.

auditory nerve	ciliary body	cochlea	cornea
eardrum	optic nerve	ossicles	
pupil	retina	semicircular canals	

The table has descriptions of parts of the eye and ear.

Complete the table by matching the descriptions, using parts from the box.

(7)

Description	Part
contains light sensitive cells	
vibrates when receives sound waves	
helps the body balance	
controls the amount of light entering the eye	
transfers vibrations across the middle ear	
turns vibrations into nerve impulses	
refracts light entering the eye	

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14. Question 1 · part a · 7 marks

November 2023 Paper 2

Further practice · Attempt part a only.

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(b) Explain why a person using both eyes finds it easier to catch a ball than a person who is blind in one eye.

(3)

(Total for Question 1 = 10 marks)

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14. Question 1 · part a · 7 marks · Working and answer

Working and answer

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